

Nighttime PM_{2.5} Concentration Estimation Based on NPP-VIIRS and Interpretable Machine Learning: The Case of Beijing–Tianjin–Hebei Region

Tong Li , Bo Li, Zhihua Han, Wenhao Zhang , and Xiufeng Yang

Abstract—Air pollution and public health issues caused by fine particulate matter (PM_{2.5}) are becoming increasingly severe. Although well-established satellite methods exist for retrieving daytime PM_{2.5} concentrations, these methods are limited by weak nighttime light radiation. To resolve these challenges, this study proposed a nighttime PM_{2.5} concentration estimation method based on explainable machine learning and low-light data. Owing to the complexity of nighttime light sources, primarily composed of artificial lighting and moonlight, both types of light were considered by simulating lunar irradiance and artificial light radiance. This study utilized nighttime lighting, meteorological, various geospatial auxiliary, simulated nighttime light radiation, and ground-based PM_{2.5} monitoring data to construct a dataset with an effective sample size of 24,311. A deep neural network model was trained to estimate nighttime PM_{2.5} concentrations. The experimental results show that, after adding the simulated nighttime light radiation, the tenfold cross-validation R^2 of the model improved from 0.6 to 0.73. In addition, 74% of site-based tenfold cross-validation R^2 values exceeded 0.7, indicating the model's robust spatial adaptability. The model was then used to estimate nighttime PM_{2.5} concentrations in the study area for 2021. The Shapley additive explanation model was applied to analyze the effect curves of different predictors on nighttime PM_{2.5} to examine the contributions of various factors. This study can serve as a reference for similar research in the future, and the proposed retrieval method offers a broad coverage of nighttime PM_{2.5} data, providing a useful supplement to ground station measurements.

Index Terms—Nighttime lighting data, nighttime particulate matter (PM_{2.5}), satellite observation, Shapley additive explanation (SHAP).

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The authors are with the School of Remote Sensing and Information Engineering, North China Institute of Aerospace Engineering, Langfang 065000, China, also with the Hebei Collaborative Innovation Center for Aerospace Remote Sensing Information Processing and Application, Langfang 065000, China (e-mail: litong@stumail.nciae.edu.cn; libo@stumail.nciae.edu.cn; hanzhihua@stumail.nciae.edu.cn; zhangwh@radi.ac.cn; yangxf_hhyg@nciae.edu.cn).

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I. INTRODUCTION

FINE particulate matter (PM_{2.5}), defined as airborne particles measuring 2.5 μm or smaller, can reach deep lung tissues and eventually enter the bloodstream [1], [2], posing a significant threat to human health [3]. Research has indicated that prolonged exposure to PM_{2.5} pollution increases mortality due to natural causes [4]. Furthermore, PM_{2.5} has profound effects on the natural environment. These fine particles can scatter and absorb solar radiation, altering the atmospheric radiative balance and affecting the Earth's climate system [5], [6]. PM_{2.5} is also a major cause of reduced atmospheric visibility [7], leading to frequent occurrences of urban haze that severely affect daily life and travel safety. China's rapid economic growth and urbanization have intensified environmental challenges, with air pollution becoming a major public concern [8]. Therefore, accurately monitoring PM_{2.5} mass concentrations and understanding its distribution is critically important [9].

Advancements in satellite remote sensing have made estimating PM_{2.5} concentrations through aerosol optical depth (AOD) increasingly common [10], [11]. Due to only faint solar radiation at night, mature AOD products are unavailable because they are limited to daytime use [12]. Currently, nighttime PM_{2.5} monitoring is primarily conducted by ground-monitoring stations, which cannot provide extensive spatially continuous PM_{2.5} monitoring [13]. The absence of stable light sources at night complicates traditional satellite-based methods for estimating nighttime PM_{2.5} [14]. Nevertheless, fine atmospheric PM_{2.5} changes often exhibit diurnal cycle characteristics [15]. The spatiotemporal distribution information of nighttime fine PM_{2.5} obtained by satellite remote sensing can fill the gap left by discrete ground measurements and be combined with daytime monitoring data to analyze diurnal variation patterns [16], [17]. In addition, the formation factors of PM_{2.5} differ significantly between day and night, with distinct sources and meteorological conditions influencing their concentrations [18], [19], [20].

Nighttime light sources are primarily comprised of artificial ground lighting and moonlight [21]. With the launch of a series of low-light remote sensing satellites, the detection of low-light signals at night has improved [22]. Due to the influence of PM_{2.5} concentration on radiance signals detected by low-light satellites at night [23], estimating nighttime PM_{2.5} concentrations using low-light data has attracted increasing attention from researchers. Several researchers have begun

developing simple linear regression models between nighttime low-light data and $PM_{2.5}$ concentrations based on the principle of aerosol extinction at night. Wang et al. [24] conducted a study in Atlanta. They found that the day–night band (DNB) of the visible infrared imaging radiometer suite (VIIRS) on the Suomi National Polar-Orbiting Partnership (NPP) satellite was highly sensitive to atmospheric aerosol variations. A multiple linear regression (MLR) model for nighttime $PM_{2.5}$ concentrations was developed using VIIRS/DNB and meteorological data, demonstrating that changes in nighttime light intensity can reflect $PM_{2.5}$ concentrations. This demonstrates the potential of low-light data for estimating nighttime $PM_{2.5}$ concentrations. A similar study in the Pearl River Delta [25] incorporated additional meteorological and topographical factors to confirm the relationship between nighttime lighting and $PM_{2.5}$ concentrations. Linear models struggle to capture the complex relationships in satellite data, whereas machine learning can address these nonlinear issues and significantly improve estimation accuracy [26]. Various machine-learning models have been applied to estimate nighttime $PM_{2.5}$ concentrations. Chen et al. [27] applied five machine-learning and MLR models to estimate nighttime $PM_{2.5}$ in Nanjing from September–December 2020. Their results showed that machine-learning models were more accurate than the MLR model. Zhao et al. [28] modeled the relationships among VIIRS/DNB, meteorological factors, and $PM_{2.5}$ in Beijing (March–May 2015) using a backpropagation neural network, emphasizing the role of meteorological variables. However, these studies were short term and considered moonlight as stray light, using only VIIRS/DNB data when the lunar phase angle was greater than 120° . This limited the accuracy of nighttime $PM_{2.5}$, which was estimated using weaker ground lighting, restricting the spatiotemporal applicability of the model. In addition, Ma et al. [29] analyzed the lunar phase angle applicability and found that using data across all lunar phases maintained accuracy while increasing the available data by 60%. Nevertheless, several challenges remain regarding nighttime $PM_{2.5}$ estimation.

- 1) The moon phase angle only partially reflects changes in lunar irradiance. More precise lunar irradiance measurements are required to better capture the effects of moonlight.
- 2) Previous studies have assumed stable ground lighting throughout the study period, overlooking its variability.
- 3) While machine learning captures complex nonlinearities in air pollution estimation, its predictions often lack transparency.

The study aimed to address the challenges above by developing a nighttime $PM_{2.5}$ estimation model using NPP/VIIRS satellite, meteorological, various geospatial auxiliary, and simulated nighttime light radiation data. The model validated the positive impact of incorporating simulated lunar irradiance and artificial lighting on nighttime $PM_{2.5}$ retrieval accuracy. The model performance was assessed through cross-validation, and the model was subsequently used to estimate the spatial distribution of nighttime $PM_{2.5}$ concentrations in the study area for 2021. In addition, Shapley's additive explanation (SHAP) was employed to analyze the contributions of individual features,

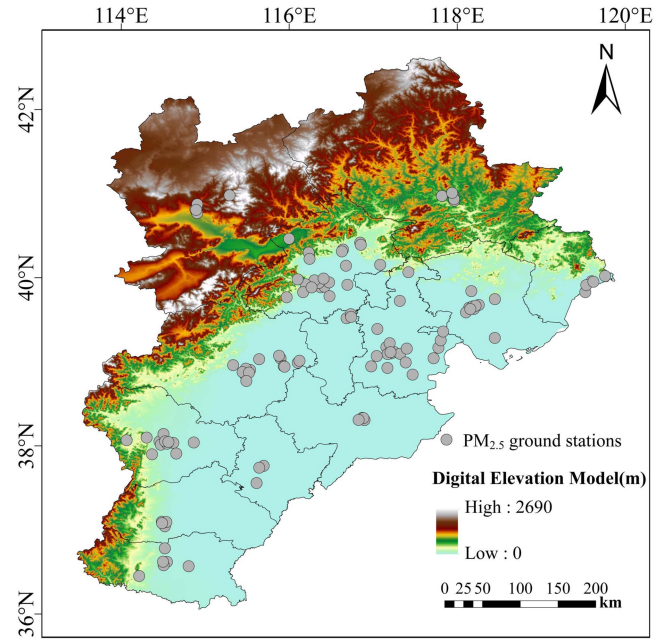


Fig. 1. Distribution of air quality testing stations in the BTH region.

addressing the interpretability in traditional machine-learning models.

II. STUDY AREA AND DATA

A. Ground $PM_{2.5}$ Data

The Beijing–Tianjin–Hebei (BTH) region is a major urban agglomeration in northern China [30] comprising the Beijing, Tianjin, and Hebei provinces. In 2021, the BTH region had 117 ground-monitoring stations (see Fig. 1). Ground-monitoring stations typically measure $PM_{2.5}$ concentrations using the beta ray attenuation or tapered element oscillating microbalance methods, with an accuracy of approximately $\pm 1.5 \mu\text{g}/\text{m}^3$ for hourly data, reflecting actual pollution levels [31]. Hourly $PM_{2.5}$ concentration data for 2021 were obtained from the China National Environmental Monitoring Station website.¹ $PM_{2.5}$ concentration values below the detection limit of $0 \mu\text{g}/\text{m}^3$ were excluded from the study [32]. The ground observation data closest in time to the satellite imagery were selected using station data recorded within 30 min of the nighttime low-light imagery acquisition.

B. NPP/VIIRS Data

The NPP satellite is an Earth observation project conducted by NASA, NOAA, and the U.S. Department of Defense. It was launched in 2011 and equipped with various remote sensing instruments, such as an infrared imager and an atmospheric vertical sounder. The VIIRS, mounted on the NPP satellite, is a multispectral remote sensing instrument capable of capturing visible and infrared data. VIIRS/DNB has a wide spectral bandpass, nominally between 500 and 900 nm, with a dynamic radiance

¹[Online]. Available: <http://www.cnemc.cn/>

range of approximately seven orders of magnitude [33]. Based on the VIIRS/DNB band data, NASA developed a black marble product suite (VNP46) that includes daily nighttime light for long-term monitoring [34]. This study used the suite's daily top-of-atmosphere nighttime radiance product (VNP46A1). This product has a spatial resolution of approximately 500 m and includes 26 scientific datasets. The VNP46A1 product removes the influence of vegetation and terrain on DNB band radiance [35]. This study used the DNB radiance, cloud mask flags, satellite zenith angle, lunar zenith angle, lunar azimuth angle, and moon phase angle from the VNP46A1 product.

C. Meteorological Data

ERA5, developed by the European Centre for Medium-Range Weather Forecasts, is a comprehensive global reanalysis dataset. It integrates various observational data, providing atmospheric information with high spatiotemporal resolutions. The data sources for ERA5 include satellite remote sensing, ground observation stations, and aircraft soundings, which cover various meteorological elements, such as temperature, cloud cover, and atmospheric pressure [36]. These data play a crucial role in meteorological research and climate analysis. ERA5 provides data with hourly time steps and a spatial resolution of 0.25°, offering crucial support for air quality monitoring and pollution source analysis. The authors in [37] and [38] showed that incorporating ERA5 data into PM_{2.5} estimate models can significantly enhance prediction accuracy and robustness. Therefore, hourly data from ERA5, including the boundary layer height (BLH), surface pressure (SP), 2-m temperature (T2M), 2-m dew point temperature (D2M), relative humidity (RH), 10-m u-component of wind (U10), and 10-m v-component of wind (V10), were used.

D. Other Geospatial Auxiliary Data

The geospatial auxiliary data used in this study included the normalized difference vegetation index (NDVI), digital elevation model (DEM), and population density (PD) data. The NDVI with a spatial resolution of 500 m was obtained from the MOD13A1 product. DEM data with a spatial resolution of approximately 900 m were obtained from GMTED2010. PD data with a 1 km resolution were derived from the LandScan dataset. These factors have been shown to influence PM_{2.5} concentrations [39], [40], [41]. To ensure consistency and accuracy, all data were resampled to match the spatial resolution of the nighttime remote sensing imagery. PD and elevation did not change significantly over the long term. The NDVI data also showed minimal variation over the 16 day period [42]. Therefore, these datasets are well suited as spatial geographic auxiliary data for estimating nighttime PM_{2.5} concentrations.

E. Simulated Nighttime Light Radiation

The MT2009 model for top-of-atmosphere spectral lunar irradiance, developed by Miller and Turner [43], was applied to calculate lunar irradiance corresponding to the satellite overpass time. The formula for calculating lunar irradiance using this

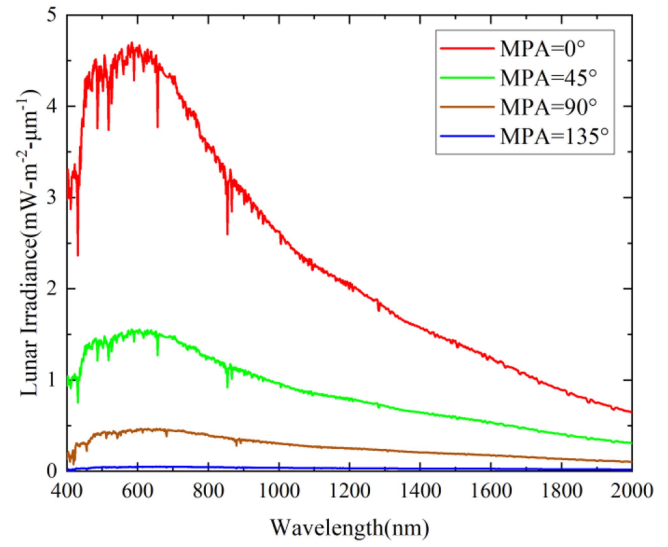


Fig. 2. Lunar irradiance simulated by MT2009 across different moon phase angles.

model is given as follows:

$$E_{mt} = L_m \Omega_m = \alpha E_0 \left(\frac{\overline{R_{se}}}{R_{sm}} \right)^2 \left(\frac{r_m}{R_{me} - r_e} \right)^2 f(\theta_p) \quad (1)$$

where α is spatially averaged lunar spectral albedo, E_0 is the solar spectral irradiance, E_{mt} is the lunar irradiance, L_m is the lunar reflected radiance, Ω_m is the apparent lunar solid angle as observed from Earth, $\overline{R_{se}}$ is the average distance between the sun and Earth, R_{sm} is the average distance between the sun and moon, r_m is the radius of the moon, R_{me} is the average distance between the Earth and the moon, r_e is the radius of the Earth, θ_p is the moon phase angle, and $f(\theta_p)$ is the moon phase function.

Fig. 2 illustrates lunar irradiance under different moon phase angles. The MT2009 model was based on a lunar spectral reflectance dataset observed by multiple scientists. In this study, lunar irradiance was simulated using the lookup tables provided by the model, which accounted for the average Earth–Sun and Earth–Moon distances. Simulated lunar irradiance was then incorporated into the model as a predictor variable.

A monthly background synthesis approach was applied to derive artificial light radiance. Previous studies demonstrated that the true radiation value of artificial light can be effectively obtained using a background synthesis method [44]. Here are the steps for constructing the background data. First, to eliminate the influence of moonlight, since the artificial light radiance detected by VIIRS/DNB can be affected by moonlight, especially during bright moon phases, it is essential to reduce this interference to obtain the accurate artificial light radiance data. Therefore, nighttime low-light images with a moon phase angle greater than 150° were selected each month. This criterion ensures that the influence of moonlight on the detected radiance is minimized. Second, as the radiance detected by the satellite can fluctuate due to varying atmospheric conditions during the radiation transmission process—such as the presence of clouds, aerosols, or varying humidity levels—it is crucial to account for

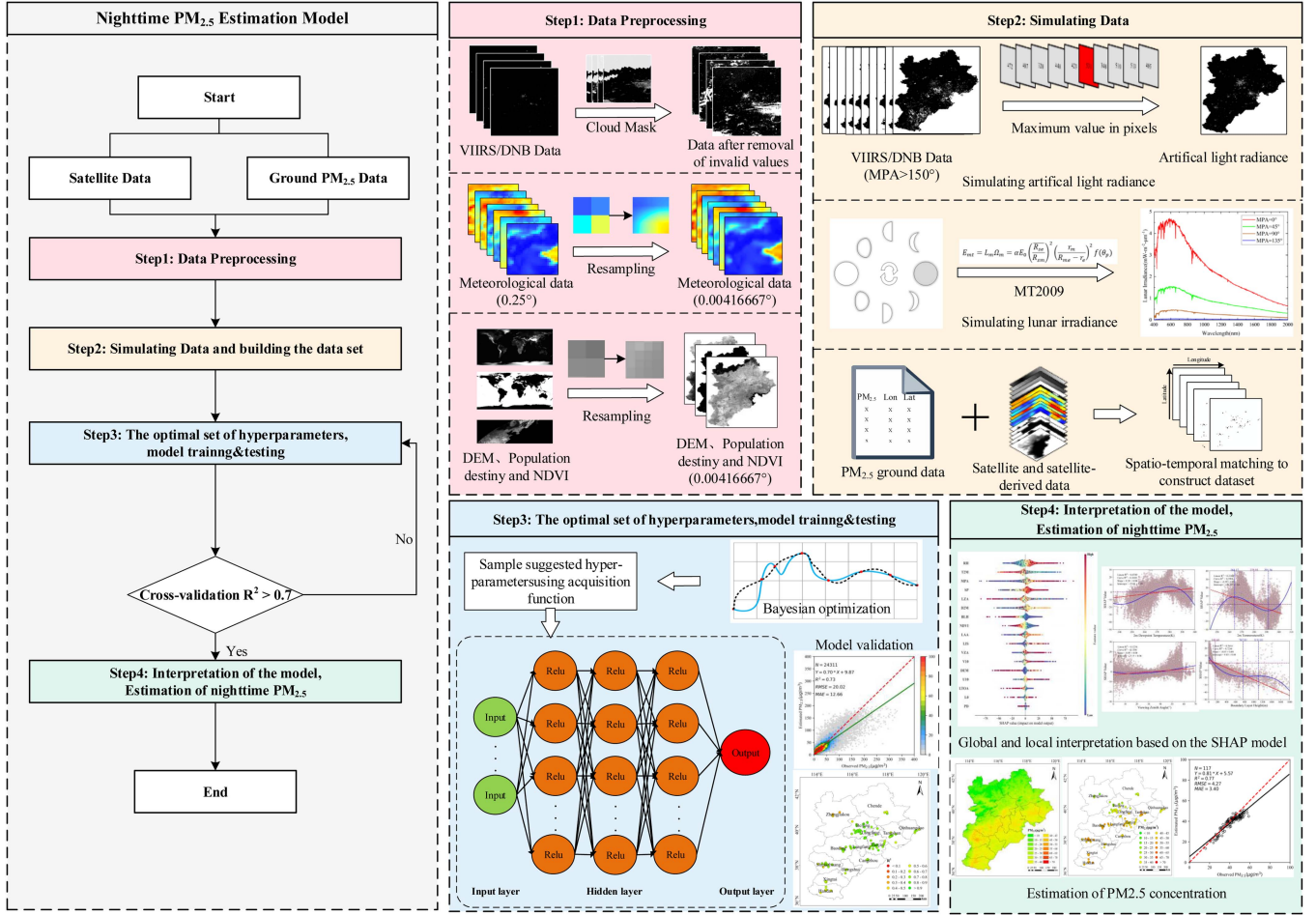


Fig. 3. Technical flowchart.

these factors to improve data reliability. The maximum radiance value within each pixel across multiple days in a month was used to achieve this. This approach assumes that the highest radiance observed is least affected by transient atmospheric disturbances. By integrating these measures, the synthesized monthly background data more accurately represent the true ground-level artificial light radiance.

III. METHODOLOGY

Fig. 3 illustrates estimating nighttime PM_{2.5} concentrations in the BTH region using low-light data. First, data preprocessing was performed. Cloud masks were applied to the VIIRS/DNB data to retain only valid observations, whereas other datasets were resampled to match the spatial resolution (0.00416667°) of the VIIRS/DNB data; artificial light radiation and lunar irradiance were simulated. VIIRS/DNB data with lunar phase angles greater than 150° were selected to construct artificial light radiance. In addition, lunar irradiance was simulated at different lunar phase angles using a lunar irradiance model (MT2009). After simulating the data, spatiotemporal matching was performed based on the location of ground stations and the overpass times of VIIRS satellites to construct the final dataset, and the Bayesian optimization method was applied to

determine the optimal hyperparameters for the deep neural network (DNN) model. The model was trained and tested using the prepared dataset. The model performance was validated using a tenfold cross-validation to ensure robustness and accuracy. Finally, SHAP analysis provided the model's global and local interpretability. This analysis helps improve our understanding of the contribution of different predictor variables to nighttime PM_{2.5} estimation. The trained model was then used to estimate the spatial distribution of nighttime PM_{2.5} concentrations across the study area.

A. Nighttime Atmospheric Radiative Transfer

The theoretical basis for estimating PM_{2.5} concentrations lies in atmospheric radiative transfer. Based on the nighttime atmospheric radiative transfer process, this study selected key parameters for the nighttime PM_{2.5} concentration estimation model. Owing to atmospheric effects, nighttime low light diminishes during transmission. Specifically, lunar irradiance from the top of the atmosphere undergoes atmospheric attenuation before reaching Earth's surface. The irradiance reflected by the surface, along with emissions from artificial light radiance, is further affected by the atmosphere before it is received and recorded by satellite sensors (see Fig. 4).

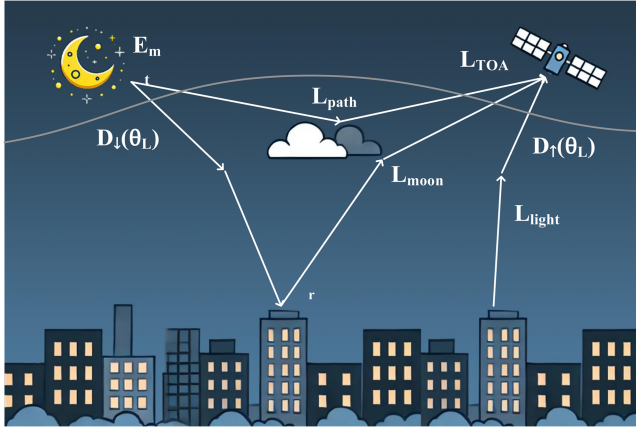


Fig. 4. Schematic diagram of the nighttime radiative transfer.

The radiance detected by the DNB can be divided into path and ground radiance, which are attenuated by the atmosphere. According to Beer's law, the nighttime low-light radiance reaching the satellite L_{TOA} can be expressed as follows:

$$L_{TOA} = L_g \cdot D_{\uparrow}(\theta_s) + L_{path} \quad (2)$$

$$D_{\uparrow}(\theta_L) = e^{-\frac{\tau}{\mu_s}} \quad (3)$$

where L_{path} is the path radiance, L_{TOA} is the radiance received by the DNB, L_g is the radiance from the ground, $D_{\uparrow}(\theta_s)$ is the atmospheric uplink transmittance, τ is the total aerosol depth, and μ_s is the cosine of the satellite zenith angle.

The radiance from the ground can be expressed as the radiance of moonlight reflected by the ground [45] L_{moon} and the radiance from artificial light L_{light}

$$L_g = L_{moon} + L_{light} \quad (4)$$

When the lunar radiance reaches the surface, it is reflected by the ground and absorbed. The reflected radiance enters the atmosphere after continuous scattering between the surface and atmospheric boundary [46]. L_{moon} can be expressed as (6), and L_{light} can be obtained through the background synthesis method

$$D_{\downarrow}(\theta_L) = e^{-\frac{\tau}{\mu_L}} \quad (5)$$

$$L_{moon} = \mu_L E_{mt} D_{\downarrow}(\theta_L) \frac{r}{\pi(1-ar)} \quad (6)$$

where μ_L is the cosine of the lunar zenith angle, $D_{\downarrow}(\theta_L)$ is the atmospheric downward transmittance, and θ_L is the lunar zenith angle. In addition, a denotes the albedo of the atmospheric hemisphere, and r represents the surface reflectance.

Finally, L_{TOA} can be articulated as follows:

$$L_{TOA} = \frac{\mu_L E_{mt} D_{\downarrow}(\theta_L) r D_{\uparrow}(\theta_s)}{\pi(1-ar)} + L_{light} D_{\uparrow}(\theta_s) + L_{path} \quad (7)$$

Through the above nighttime radiative transfer analysis, it can be seen that there is a certain connection between L_{TOA} and AOD. Building upon the radiative transfer framework, the relationship between the vertically measured AOD τ and PM_{2.5} concentration near the surface has been further explored. According to Wang and Christopher's study [47], this relationship

can be expressed as follows:

$$\tau = PM_{2.5} f(RH) Q \cdot BLH \quad (8)$$

where $f(RH)$ represents the hygroscopic growth factor of particles, Q denotes the mass extinction efficiency, and BLH refers to the effective BLH.

At night, due to the complex relationship between nighttime lighting and PM_{2.5} and the uncertainty in obtaining these parameters under nighttime conditions, it is impossible to accurately calculate PM_{2.5} concentrations using theoretical models. Therefore, this article uses machine learning to model these nonlinear relationships.

B. DNN Model

DNN is a type of neural network containing multiple hidden layers. It stimulates the connections between neurons to address complex regression problems. In each layer, the neurons nonlinearly transform the input data using activation functions and pass the results to the next layer. By training many samples, the DNN model can capture complex features within the data, thereby enabling efficient input data modeling. DNN models can automatically capture and extract high-level features from the input data, leading to greater accuracy when estimating atmospheric parameters. This study employed a DNN model to simulate nighttime PM_{2.5} concentrations. Finally, PM_{2.5} can be expressed as follows:

$$PM_{2.5} = \text{DNN} \left(\begin{array}{c} L_{TOA}, L_0, \\ LAA, LZA, VZA, MPA, LIS \\ BLH, D2M, RH, SP, T2M, U10, V10, \\ NDVI, DEM, PD \end{array} \right) \quad (9)$$

where L_0 is the artificial optical radiation obtained by the background field method simulation, LAA is the lunar azimuth angle, LZA is the lunar zenith angle, VZA is the sensor zenith angle, MPA is the lunar phase angle, and LIS is the lunar irradiance obtained by the MT2009 lunar irradiance model simulation [E_{mt} in (1)].

C. SHAP Model

The SHAP model is widely used in machine learning to interpret complex model outputs. It leverages cooperative game theory to allocate the contribution of each feature to the model predictions fairly [48]. In this context, SHAP calculates the values for each feature to explain the model's output and measure its contribution to the predictions. The SHAP values were derived using the following:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (M - |S| - 1)!}{M!} [f(S \cup \{i\}) - f(S)] \quad (10)$$

where N represents the set of all features, M is the number of features, S is a subset of N , $f(S)$ is the output based on S , and $f(S \cup \{i\})$ is the output when feature i is added to S .

This study employed the SHAP model to reveal the marginal contributions of various predictors of nighttime PM_{2.5} concentrations. Fig. 5 presents a local perspective for an individual

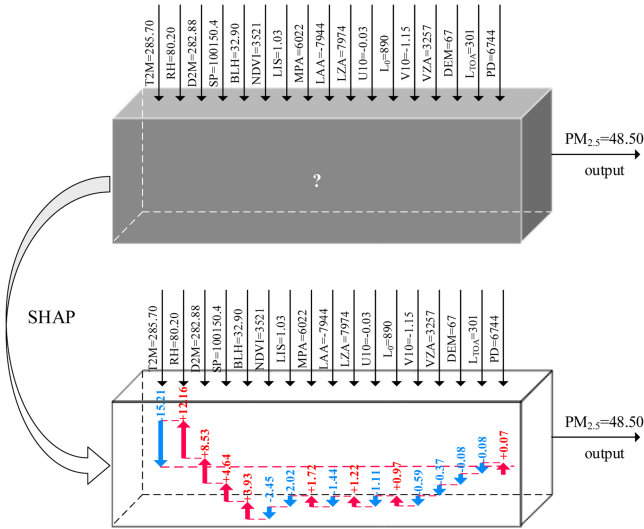


Fig. 5. Perspective view of Shapley analysis of a single sample.

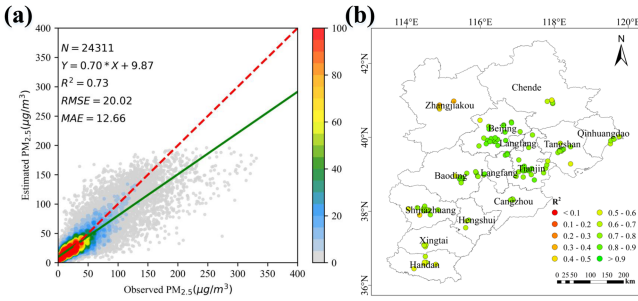


Fig. 6. Model tenfold cross-validation accuracy. (a) Cross-validation accuracy based on samples. (b) Cross-validation accuracy based on stations.

sample, showing the extent to which each predictor variable drives the model output from the base value (mean prediction). The observed $\text{PM}_{2.5}$ concentration at the station is $49 \mu\text{g}/\text{m}^3$, whereas the estimated concentration is $48.5 \mu\text{g}/\text{m}^3$. In addition to achieving a high-accuracy fit, the transparency of the model was greatly enhanced, allowing us to precisely pinpoint and compare the influences of these factors.

IV. RESULTS AND DISCUSSION

A. Model Performance

Fig. 6 illustrates the validation of the model. Fig. 6(a) illustrates the tenfold cross-validation results between the estimated and observed $\text{PM}_{2.5}$ concentrations based on the sample data; the results show R^2 of 0.73, root-mean-square error (RMSE) of $20.02 \mu\text{g}/\text{m}^3$, and mean absolute error (MAE) of $12.66 \mu\text{g}/\text{m}^3$ after training the DNN model, indicating that the model performs well in the BTH region. Fig. 6(b) shows the station-based tenfold cross-validation results for the study period. To further validate the spatial applicability of the model, the 117 stations were divided into ten subsets for cross validation. Most R^2 values from the station-based validation exceeded 0.7, accounting for approximately 74% of the total, indicating the model's robust spatial adaptability. These values highlight the ability of the

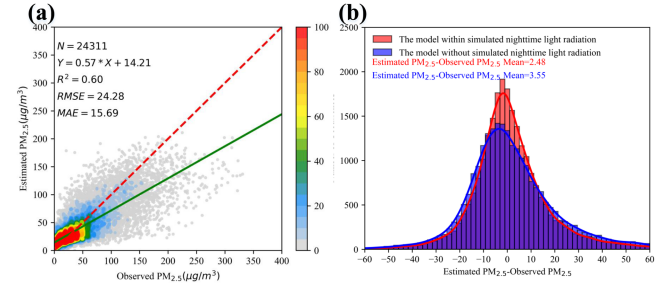


Fig. 7. (a) Tenfold cross-validation accuracy of the model without considering lunar irradiance and artificial ground radiance. (b) Comparison of average errors of two models.

model to adapt to various geographical and environmental conditions.

B. Impact of Lunar Irradiance and Artificial Ground Radiance on the Model

A model was constructed to assess the impact of simulated nighttime light radiation on model predictions without incorporating both features to simulate nighttime $\text{PM}_{2.5}$ concentrations. Fig. 7 shows the validation of the model without considering the simulated nighttime light radiation. In Fig. 7(a), compared with the model using simulated nighttime light radiation, the tenfold cross-validation R^2 dropped from 0.73 to 0.6, whereas the RMSE and MAE increased from 20.02 and $12.66 \mu\text{g}/\text{m}^3$ to 24.28 and $15.69 \mu\text{g}/\text{m}^3$. Fig. 7(b) shows the average relative errors of the two models. The average error for the model without simulated nighttime light radiation was $3.55 \mu\text{g}/\text{m}^3$, and the average error for the model within simulated nighttime light radiation was $2.48 \mu\text{g}/\text{m}^3$. Moreover, the average error distribution of the model considering simulated nighttime light radiation is more concentrated around $0 \mu\text{g}/\text{m}^3$, whereas the error of the other model is larger.

This study also compared the model with those using only nighttime light data in previous research. The model demonstrated better stability and accuracy than Ma et al.'s study [49], where the R^2 ranged from 0.51 to 0.63 and RMSE ranged from $54.25 \mu\text{g}/\text{m}^3$ to $14.11 \mu\text{g}/\text{m}^3$. In contrast, studies by the authors in [25], [27], [28], and others achieved higher model accuracies, with R^2 values of 0.82, 0.81, and 0.83, respectively. However, these models only used data with lunar phase angles greater than 120° , limiting their application mainly to urban and surrounding areas. Our study offers a broader spatial coverage, providing more comprehensive nighttime $\text{PM}_{2.5}$ estimation across a larger region.

C. Impact Factor Influence Analysis

To enhance our understanding of the decision-making processes employed by the model, this study calculated the SHAP values for the 17 influencing factors used as inputs. Fig. 8 illustrates the impact of the input features on the model from a global perspective. In Fig. 8, each point represents a sample, and the colors of the points indicate the magnitudes of the feature values. The points were arranged in descending order of the

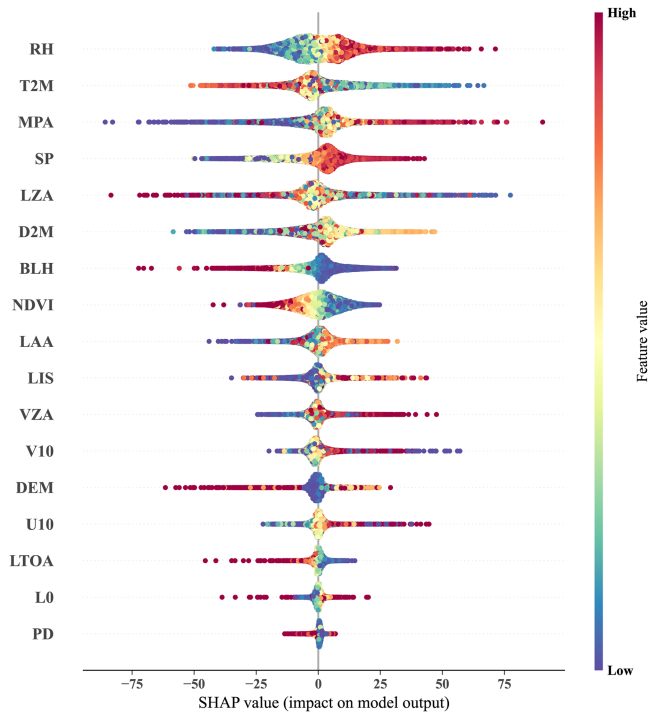


Fig. 8. Influence of input features on PM_{2.5} prediction (from global perspective).

mean absolute SHAP value, revealing the impact of each of the 17 factors on the prediction results. RH, T2M, MPA, and SP were the top four features with the greatest impact on nighttime PM_{2.5} estimation. Specifically, these features exhibited a wide range of SHAP values, with either large positive or smaller negative values, significantly driving fluctuations of the estimated PM_{2.5} concentrations. The wide range of SHAP values indicates that the factors affecting nighttime PM_{2.5} estimation are complex. Fig. 8 shows that a higher RH has a stronger positive impact on the model mainly because RH promotes hygroscopic growth of aerosols, leading to increased particle size and scattering efficiency [50], thereby increasing the mass concentration of PM_{2.5} [51]. In contrast, higher temperatures facilitate the deepening of the atmospheric mixing layer, enhancing vertical dispersion, which has a negative impact on the estimation results [52]. Furthermore, seasonal temperature variations can further influence emissions, as colder periods often lead to increased combustion-related emissions due to heating demands. Meanwhile, higher pressure usually indicates more stable atmospheric conditions, making it difficult for pollutants to disperse. In addition, a greater BLH facilitated the dispersion of nighttime PM_{2.5} [53]. When the NDVI is higher, it has a negative contribution to nighttime PM_{2.5} concentrations, as greater vegetation coverage helps to purify air [54]. SHAP values allow for the measurement of each influencing factor in contributing to the model output [55], a task that cannot be accomplished by the coefficient of determination or feature importance rankings, as these only indicate predictive power [56].

Fig. 9 shows the influence curves of each feature on the estimated nighttime PM_{2.5} concentrations, illustrating the varying

impacts of these variables across different value ranges. This highlights the nonlinear and complex relationships between feature variables and PM_{2.5} concentrations. Linear and cubic regression fitting were applied to analyze these relationships. For example, Fig. 9(a) indicates that artificial light radiance generally positively contributes to PM_{2.5} predictions, although the impact varies depending on the radiance range. Similarly, Fig. 9(b) shows a predominantly negative relationship between DNB radiance and PM_{2.5} concentration, associated with reduced scattering when PM_{2.5} levels are low [57]. In this study, we obtained the thresholds for reversing feature variable contributions through statistical methods. As shown in Fig. 9(c), when BLH is below 149.6 m, BLH predominantly contributes positively to PM_{2.5} concentration, this is because lower boundary layers usually mean higher pressure, which stabilizes the atmosphere and limits the dispersion of pollutants, leading to their accumulation near the surface [58]. However, when BLH exceeds 149.6 m, BLH predominantly negatively contributes to PM_{2.5} concentration. As BLH increases, it is generally accompanied by a decrease in SP, making it easier for pollutants to disperse [59]. Research has also shown that, when the boundary layer exceeds 200 m, the probability of pollution events further decreases [60], which is consistent with the results of this study. As shown in Fig. 9(e), when the T2M is below 279.91 K, the temperature in the BTH region is approximately 6.76 °C, indicating a relatively cold environment. Low temperatures are often associated with increased heating, leading to higher PM_{2.5} concentrations. When temperatures range between 264.17 and 291.36 K, the positive impact on PM_{2.5} gradually diminishes and transitions to a negative impact. On the one hand, higher temperatures reduce heating activities; on the other hand, higher temperatures improve atmospheric dispersion conditions, thereby reducing PM_{2.5} concentrations. In contrast to these complex relationships, the influences of RH and NDVI on PM_{2.5} concentrations exhibited a linear relationship, with linear fit R^2 values of 0.7580 and 0.6413, respectively. As shown in Fig. 9(f), RH is positively correlated with PM_{2.5} predictions. Low humidity has a negative impact on PM_{2.5}; however, as humidity increases, the negative impact diminishes until humidity reaches 54.73%, after which further increases in humidity lead to higher PM_{2.5} concentrations. Conversely, NDVI shows an overall negative correlation with PM_{2.5} predictions. When the NDVI exceeded 0.05, its positive impact on PM_{2.5} sharply decreased, and after exceeding 0.28, its negative impact sharply increased. Above 0.64, the negative impact on PM_{2.5} lessens. As shown in Fig. 9, the NDVI and RH exhibited linear relationships with the contribution values, whereas the other feature parameters showed nonlinear relationships. This suggests that NDVI and RH are closely associated with variations in PM_{2.5} concentrations. However, the observed relationship between NDVI and PM_{2.5} may primarily reflect seasonal patterns, as NDVI values tend to increase during warmer months when heating activities are reduced, and decrease in colder months when emissions from heating activities are higher. In contrast, the nonlinear relationships of other variables indicate that their impact on PM_{2.5} concentrations may depend on interactions with other atmospheric parameters and the complex meteorological environment.

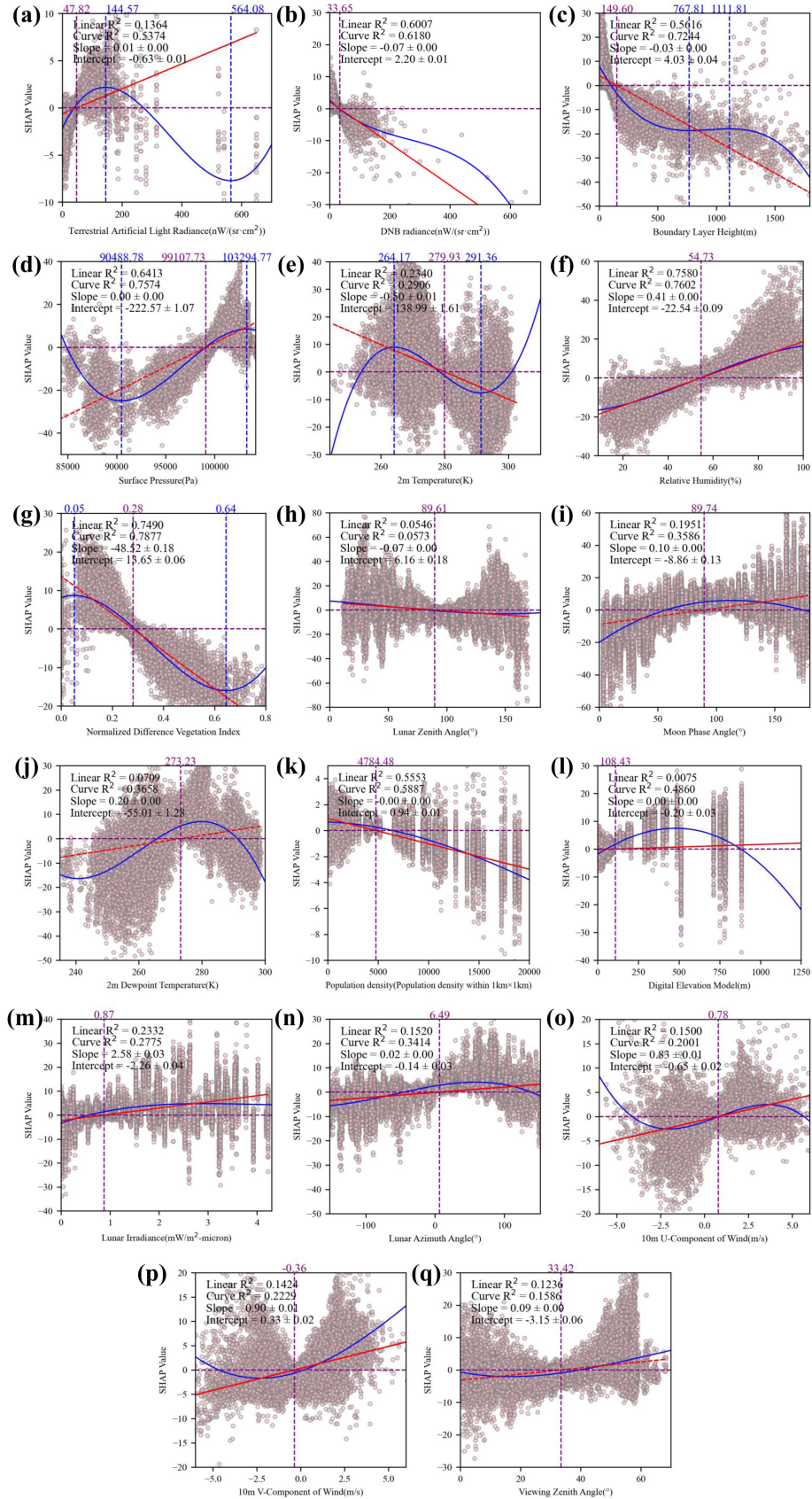


Fig. 9. Influence of individual features on estimated nighttime $PM_{2.5}$ concentrations. (a-q) Corresponding influence curves for each feature.

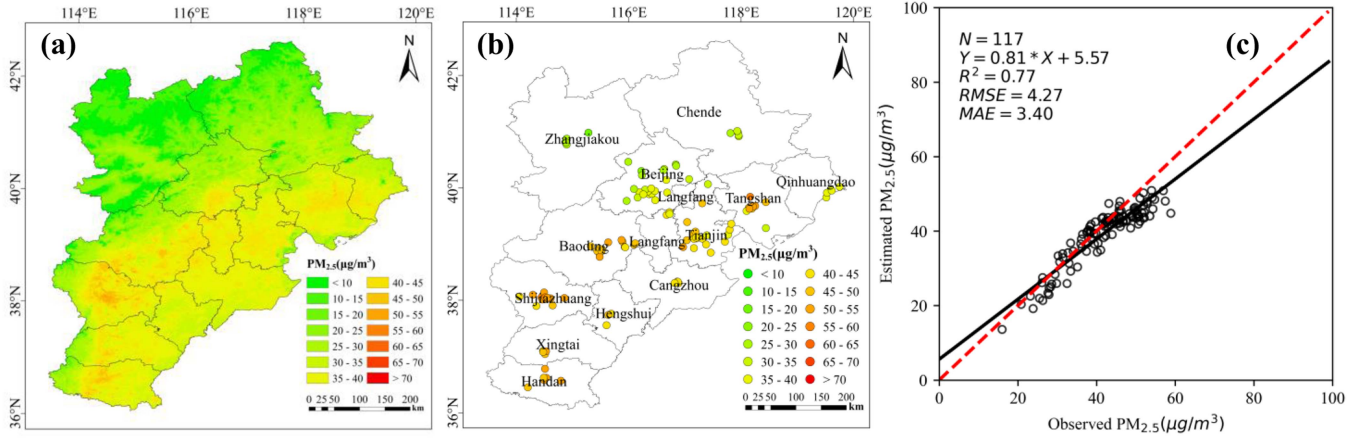


Fig. 10. (a) Annual average nighttime PM_{2.5} estimation results. (b) Annual average ground-monitoring station observation results. (c) Fitting accuracy of the estimation results with the ground stations.

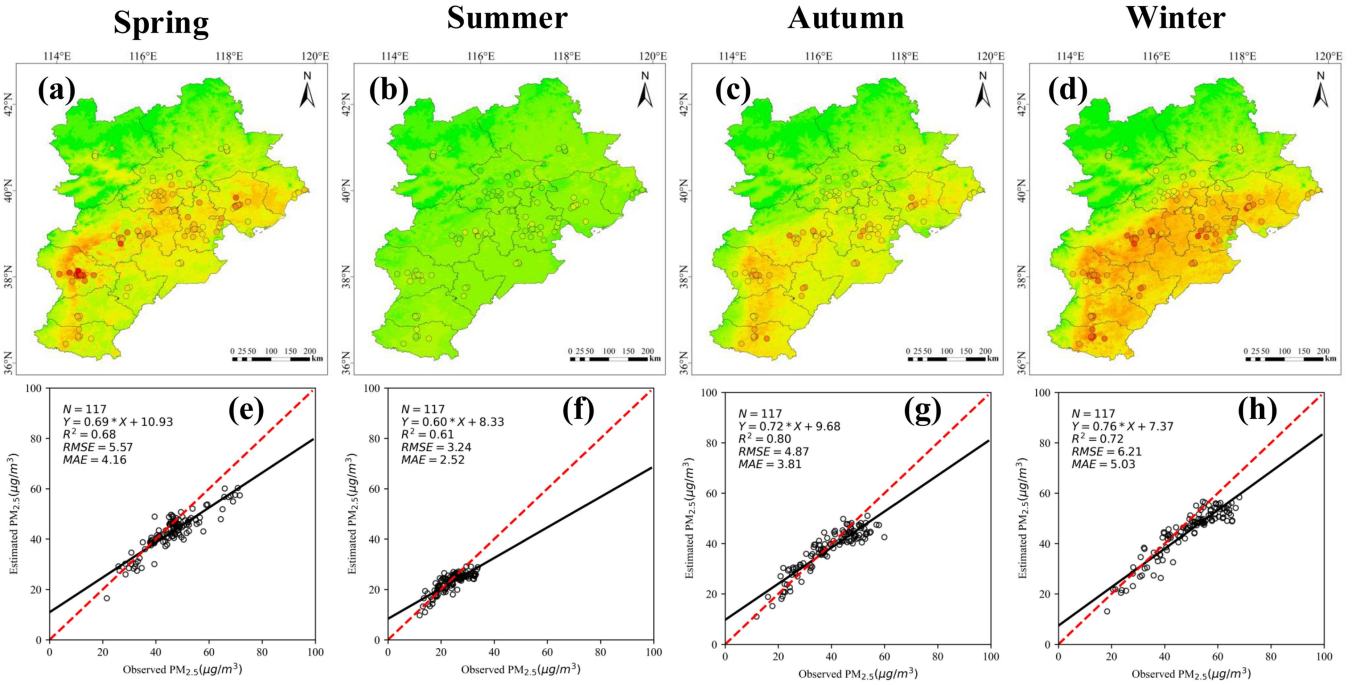


Fig. 11. Seasonal distribution of PM_{2.5}. (a)–(d) Spatial distribution of nighttime PM_{2.5} for the four seasons. (e)–(h) Estimation accuracy for different seasons.

Understanding the impact of these features on PM_{2.5} concentration can serve as a foundation for future studies. This allowed for better feature selection and a more detailed analysis of the factors affecting PM_{2.5} concentrations.

D. Spatiotemporal Distribution of PM_{2.5}

Fig. 10 shows the annual average spatial distribution of nighttime PM_{2.5} concentrations in the BTH region for 2021. The inversion results were compared with ground station measurements to verify the model's accuracy. As shown in Fig. 10(c), the estimated results are correlated with the ground station observations ($R^2 = 0.77$, $RMSE = 4.27 \mu\text{g}/\text{m}^3$, and

$MAE = 3.40 \mu\text{g}/\text{m}^3$). The spatial distribution [see Fig. 10(a)] reveals higher concentrations in the south and east, and lower concentrations in the north and west. This pattern reflects the combined influence of geographical and anthropogenic factors. For example, areas near the Taihang Mountains, with less human activity, exhibited lower PM_{2.5} concentrations, while more developed urban areas, such as Beijing and Tianjin in the central and eastern regions, had relatively higher levels [61].

Fig. 11 presents the seasonal retrieval results for nighttime PM_{2.5} concentrations. As shown in Fig. 11(e)–(h), the fitting accuracy (R^2) between the inversion results and ground station observations was above 0.6 for each season, with the highest accuracy in autumn (0.8) and the lowest in summer (0.61). These

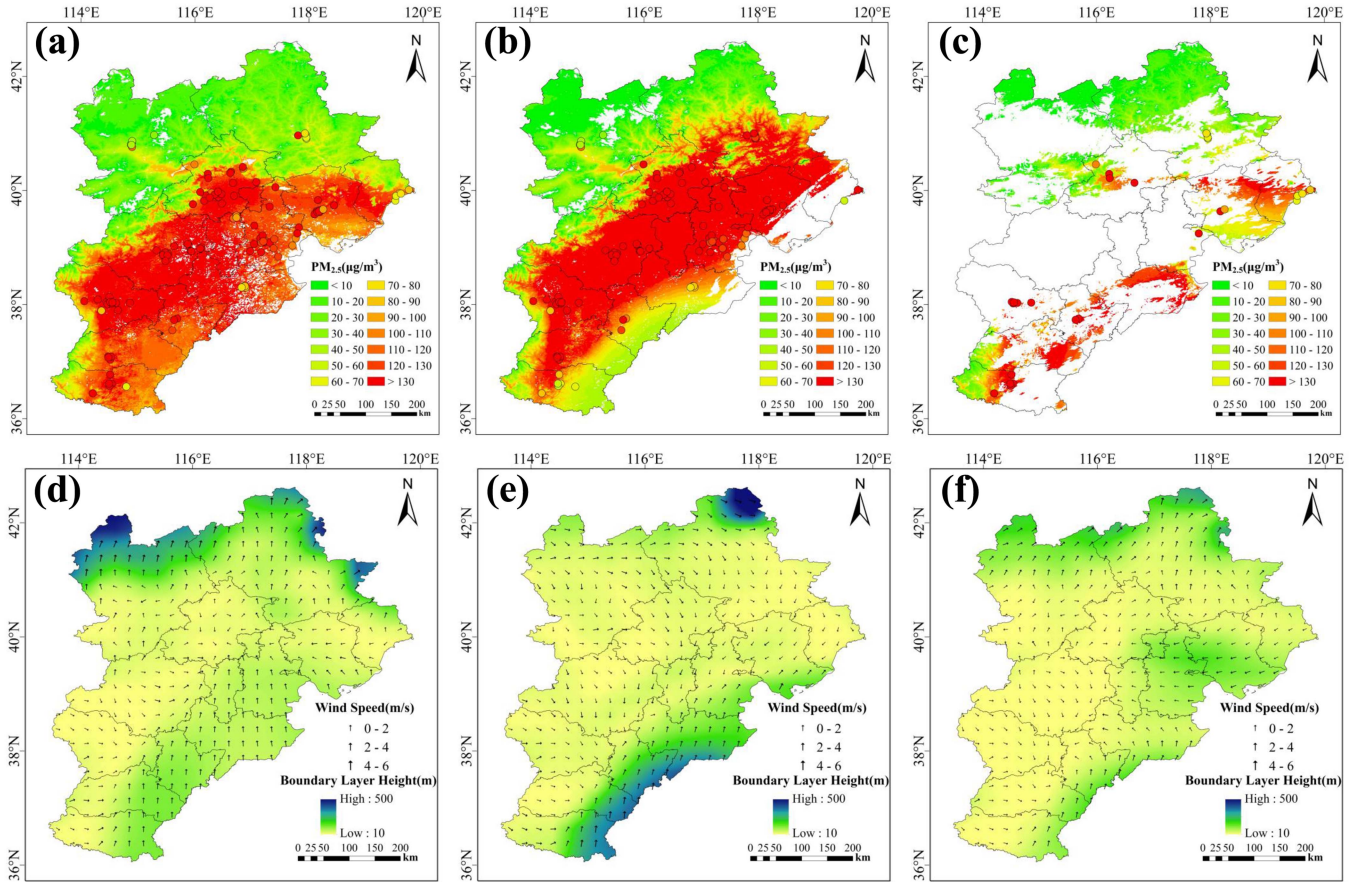


Fig. 12. (a)–(c) $PM_{2.5}$ estimation results for 10–12 February 2021. (d)–(f) BLH and wind speeds for 10–12 February 2021.

differences are partly due to varying sample sizes; the dataset contained 24 311 samples, but summer had the fewest valid samples (4467) due to high cloud cover that rendered many samples invalid. Fig. 11(a)–(d) shows the seasonal distribution of nighttime $PM_{2.5}$ concentrations, revealing that pollution is most severe in winter, followed by spring and autumn, with the lowest levels in summer. The higher pollution in winter is mainly attributed to heating activities, particularly coal burning, which increases emissions and $PM_{2.5}$ concentrations, especially at night [62]. In contrast, summer's high temperatures, strong solar radiation, and frequent rainfall promote the vertical dispersion of pollutants and remove $PM_{2.5}$ from the air, resulting in lower $PM_{2.5}$ concentrations. Meteorological conditions also vary seasonally, contributing to the observed differences in pollution levels.

V. DISCUSSION

To highlight the practical value of this study, we analyzed a pollution event during the 2021 Spring Festival. Fig. 12 illustrates a pollution event during the Spring Festival in the BTH region, with Fig. 12(a)–(c) showing the situation at 02:00 on 11 February, 03:00 on 12 February, and 02:00 on 13 February (local time). Through daily nighttime $PM_{2.5}$ inversion, the model accurately captured this pollution process. On 12 February, elevated pollutant concentrations were observed likely due to

increased heating activities during the Spring Festival [63], which led to higher $PM_{2.5}$ concentrations. At 03:00 on 12 February, the $PM_{2.5}$ concentration further increased, primarily due to traditional fireworks and firecrackers during the festival, which directly contributed to the rise in near-surface $PM_{2.5}$ concentrations [64]. As shown in Fig. 12(e), the BLH distribution pattern was similar to that of $PM_{2.5}$. Lower BLH values limited the dispersion of $PM_{2.5}$, further increasing concentrations. By 02:00 on 13 February, the satellite imagery was affected by cloud cover, resulting in the large areas of invalid data. However, as the BLH increased, the pollution levels in the eastern BTH region correspondingly decreased. This analysis of the pollution event validates the model's ability to capture dynamic pollution events, demonstrating its practical value in accurately tracking and predicting air quality fluctuations during specific periods.

Nevertheless, our study has certain limitations.

- 1) The MT2009 model has been widely used to calculate lunar irradiance; however, the model has limitations in certain specific scenarios, particularly in the accuracy of its predictions at extreme lunar phase angles (less than 5° or greater than 120°). These inaccuracies can introduce potential errors when using the model in certain lunar cycle phases [65]. Future improvements in lunar irradiance modeling, such as the use of the higher precision robotic lunar observatory model [66], could reduce the error margin

and enhance the reliability of lunar irradiance estimates in subsequent studies [67].

- 2) In this study, we used the maximum radiance value within a month for pixels with a moon phase angle greater than 150° as the artificial light background field. While this method may provide a reasonable simulation of artificial light radiance in bright pixels, it could potentially overestimate the influence of surrounding bright pixels on dark pixels due to scattering effects. In addition, this approach may be affected by transient atmospheric conditions, which can introduce some uncertainty into the simulated results. Future improvements could involve obtaining more accurate ground-level artificial light radiance data to reduce these uncertainties in the nighttime PM_{2.5} estimation process.
- 3) The VNP46A1 nighttime light product does not account for the effects of surface bidirectional reflectance distribution function (BRDF) [68] and view geometry on nighttime lighting [69]. Although this study used satellite angle data, these effects may still contribute to inaccuracies in the observed nighttime light levels. Further calibration of nighttime light products, including correction for atmospheric, BRDF, and view geometry effects, would improve the accuracy of the foundational data used in such studies. In future research, using VNP46A2 product, which has undergone atmospheric and BRDF corrections, may yield more precise nighttime lighting data for PM_{2.5} estimation.
- 4) In this study, we utilized data across all lunar phase angles to estimate nighttime PM_{2.5} concentrations in the BTH region. Specifically, in areas without artificial lighting, DNB observations at large lunar phase angles may lack sufficient information for the accurate PM_{2.5} estimation. In such cases, the potential predictive capability of the model in these regions may stem from the strong correlation between meteorological factors and PM_{2.5}. However, the absence of ground-monitoring stations in these regions hinders the effective validation of our estimates. Future studies should refer to physics-based modeling approaches to refine methodologies and improve both model applicability and estimation accuracy.

VI. CONCLUSION

This study constructed a nighttime PM_{2.5} estimation model for the BTH region using NPP/VIIRS satellite data, simulating nighttime radiance, meteorological data, and various geospatial auxiliary data. The model was validated against ground-monitoring station data. Following validation, the model was used to estimate nighttime PM_{2.5} concentrations across the region for 2021. In addition, SHAP was used to enhance the interpretability of the estimation results. The following conclusions were drawn.

- 1) By simulating and incorporating lunar irradiance and artificial ground radiance into the model, the R^2 improved from 0.6 to 0.73, with RMSE decreasing from 24.28 $\mu\text{g}/\text{m}^3$ to 20.02 $\mu\text{g}/\text{m}^3$ and MAE dropping from 15.69 $\mu\text{g}/\text{m}^3$ to

12.66 $\mu\text{g}/\text{m}^3$. This demonstrates that these two features enhance the accuracy of nighttime PM_{2.5} concentration estimates.

- 2) The global interpretability of SHAP was used to quantify the contribution of each predictor to the estimation, with RH and T2M being the two feature variables that contributed the most on average to the model's results. In addition, the study also focused on the model's local interpretability by examining the relationship between individual feature variables and their contribution values. It was found that NDVI and RH exhibited a clear linear relationship with the contribution values, with linear regression coefficients of 0.75 and 0.76, respectively. In contrast, the relationships between other predictors and their contribution values were more complex and nonlinear.
- 3) Pollution events were traced using backward trajectory analysis. The results revealed that nighttime PM_{2.5} pollution in the BTH region was particularly severe during the Spring Festival, particularly around the local time of New Year's Eve. During the three days surrounding New Year's Eve, the average nighttime PM_{2.5} concentrations in the region were 80 $\mu\text{g}/\text{m}^3$ on 10 February, 95 $\mu\text{g}/\text{m}^3$ on 11 February, and 53 $\mu\text{g}/\text{m}^3$ on 12 February, exceeding the World Health Organization's limit of 15 $\mu\text{g}/\text{m}^3$.

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Tong Li received the B.E. degree in electronic information from Henan University of Science and Technology, Henan, China, in 2023. He is currently working toward the M.S. degree in the field of electronic information with the School of Remote Sensing and Information Engineering, North China Institute of Aerospace Engineering, Langfang, China.

His research interests include atmospheric remote sensing and remote sensing image processing.



Bo Li received the B.E. degree from Hebei University of Engineering, Hebei, China, in 2023. He is currently working toward the M.S. degree in the field of electronic information at the School of Remote Sensing and Information Engineering, North China Institute of Aerospace Engineering, Langfang, China.

His research interests include atmospheric remote sensing and remote sensing image processing.



Zhihua Han received the B.E. degree from Zhengzhou University of Light Industry, Henan, China, in 2022. She is currently working toward the M.S. degree in the field of aeronautical and astronautical science and technology at the School of Remote Sensing and Information Engineering, North China Institute of Aerospace Engineering, Langfang, China.

Her research interests include atmospheric remote sensing and quantitative estimation of anthropogenic carbon emissions.



Wenhao Zhang received the B.E. degree in surveying and mapping engineering from the Hebei University of Science and Technology, Tangshan, China, in 2010, the M.S. degree in photogrammetry and remote sensing from the Chinese Academy of Surveying and Mapping, Beijing, China, in 2013, and the Ph.D. degree in signal and information processing from the Institute of Remote Sensing and Digital Earth, Chinese Academy of Sciences, Beijing, China, in 2016.

He is currently an Associate Professor with the North China Institute of Aerospace Engineering, Langfang, China. His research interests include atmospheric environment remote sensing, atmospheric aerosol, and remote sensing data processing.



Xiufeng Yang received the M.S. degree in geomatics science and technology from Henan Polytechnic University, Jiaozuo, China, in 2014.

She is currently an Associate Professor with the North China Institute of Aerospace Engineering, Langfang, China. Her research interests include surveying and mapping technology, geospatial data analysis, and remote sensing applications.